

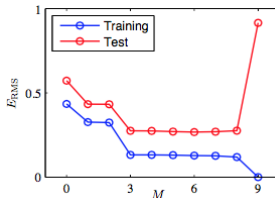


CSCE 633-700: Machine Learning

Linear regression

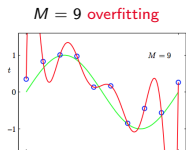
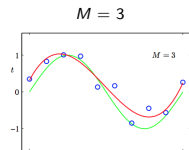
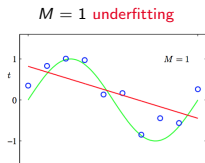
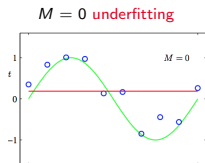
Overfitting

- The risk of using highly flexible (complicated) models without enough data
- Leads to **poor generalization**
- How to detect overfitting?
 - Plot model complexity (e.g. polynomial order) versus objective function
 - As complexity increases, performance on training improves, while on testing first improves and then deteriorates
- How to avoid overfitting?
 - More data or **regularization**

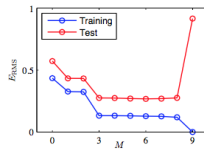


Overfitting

Example: Non-linear regression $y = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M$
Samples from a sine function $x_i = \sin(t_i)$, $t_i \sim \text{Uniform}(0, 2\pi)$



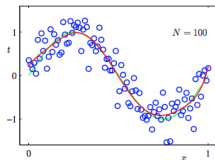
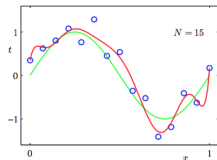
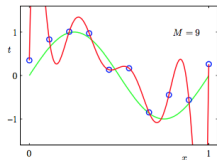
	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0	0.19	0.82	0.31	0.35
w_1		-1.27	7.99	232.37
w_2			-25.43	-5321.83
w_3			17.37	48568.31
w_4				-231639.30
w_5				640042.26
w_6				-1061800.52
w_7				1042400.18
w_8				-557682.99
w_9				125201.43



As model becomes more complex, performance on training keeps improving while on test data improve first and deteriorate later.

How can we avoid overfitting?

One solution: Use more training data



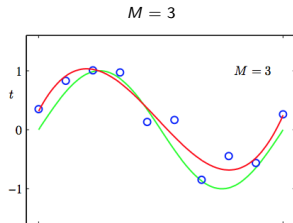
What if we don't have a lot of data?

Another solution: Use less features (e.g. feature selection algorithms)

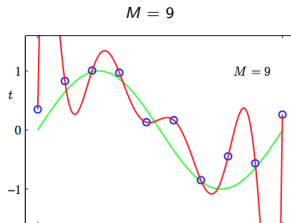
Intuitively, this will reduce the complexity of the model, therefore it is likely to result in less overfitting.

How can we avoid overfitting?

A more general solution: Regularization



$$y = w_0 + w_1x + w_2x^2$$



$$y = w_0 + w_1x + w_2x^2 + \dots + w_9x^9$$

How about penalizing and making small $w_3, \dots, w_9$?

The cost function to be minimized would become:

$$J(\mathbf{w}) = RSS(\mathbf{w}) + w_3^2 + \dots w_9^2$$

But we may not know in advance which parameters we want to penalize $\rightarrow$ So we can penalize them all

How can we avoid overfitting?

A more general solution: Regularization

Suppose we have a learning model whose evaluation criterion $EC(\mathbf{w})$ we want to optimize with respect to weights $\mathbf{w} = [w_1, \dots, w_D]^T$

- $J(\mathbf{w}) = EC(\mathbf{w}) + \lambda \sum_{d=1}^D w_d^2 = EC(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$

→ l2-norm regularization

- $J(\mathbf{w}) = EC(\mathbf{w}) + \frac{\lambda}{N} \sum_{d=1}^D w_d^2$

(as #data N increases, we need to worry less about overfitting)

- $J(\mathbf{w}) = EC(\mathbf{w}) + \lambda \sum_{d=1}^D \|w_d\| = EC(\mathbf{w}) + \lambda \|\mathbf{w}\|$

→ l1-norm regularization

Evaluation criterion $EC(\mathbf{w})$ can be RSS or log-likelihood for linear regression, negative cross-entropy for logistic regression, etc.

$\lambda \geq 0$ is the model complexity penalty

Regularization for Non-Linear Regression

ℓ_2 -norm regularization

Linear: $J(\mathbf{w}) = \text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2 = (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$

Non-linear: $J(\mathbf{w}) = \text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2 = (\mathbf{y} - \Phi\mathbf{w})^T (\mathbf{y} - \Phi\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$

Closed-form solution:

Linear: $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}_{D \times D})^{-1} \mathbf{X}^T \mathbf{y}$

Non-linear: $\mathbf{w}^* = (\Phi^T \Phi + \lambda \mathbf{I}_{D \times D})^{-1} \Phi^T \mathbf{y}$

The above reduces to ordinary least squares (OLS) solution when $\lambda = 0$
(see handout for derivation)

Overfitting

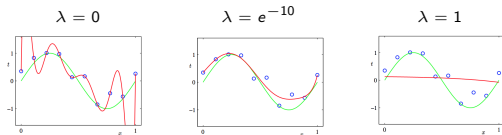
Why would we need regularization in linear regression?

- Multicollinearity (or collinearity): the existence of near-linear relationships among the features
 - e.g., same variable represented in two different units
 - e.g., three ingredients of a mixture summing to 100% ($x_1 + x_2 + x_3 = 100$)
- Multicollinear variables result in data matrices close to non-invertible, therefore causing inaccurate estimates of the regression coefficients
- Regularization would cause some of the coefficients (potentially the ones corresponding to one of the multicollinear variables) to be close to zero

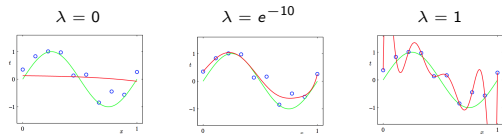
Regularization for Non-Linear Regression

Question: Assume a set of samples generated from a sine function $x_i = \sin(t_i)$ (green line), modeled with **regularized** non-linear regression $y = w_0 + w_1x + \dots + w_9x^9$. How does the resulting model (red line) look as we increase the amount of regularization λ ?

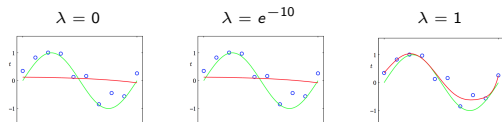
A)



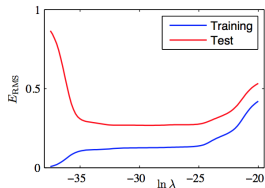
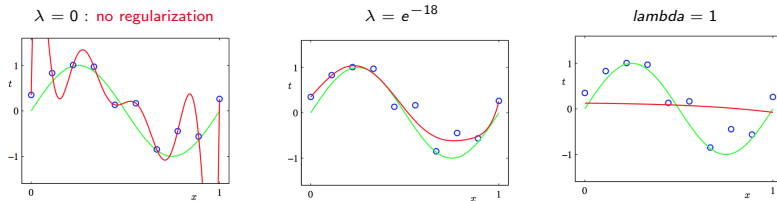
B)



C)



Regularization for Non-Linear Regression



	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
w_0	0.35	0.35	0.13
w_1	232.37	4.74	-0.05
w_2	-5321.83	-0.77	-0.06
w_3	48568.31	-31.97	-0.06
w_4	-231639.30	-3.89	-0.03
w_5	640042.26	55.28	-0.02
w_6	-1061800.52	41.32	-0.01
w_7	1042400.18	-45.95	-0.00
w_8	-557682.99	-91.53	0.00
w_9	125201.43	72.68	0.01

For a complex model ($M = 9$), training error increases with increasing regularization.